

# A First Lean Formalization Audit of the IUT III Corollary 3.12 Corridor

IUT Lean formalization project

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## Abstract

This note records the present state of a Lean formalization of a narrow part of Mochizuki’s inter-universal Teichmüller theory: the passage from IUT III, Theorem 3.11 to Corollary 3.12, with attention to the Scholze–Stix objection. The current Lean code proves conditional route theorems, obstruction lemmas, and the final ordered-real algebra in Step (xi-f). It does not prove Corollary 3.12 and does not refute it. The main formal result so far is that a supplied hull/log-volume  $q$ -to- $\Theta$  comparison gives  $C_\Theta \geq -1$ , and that a nonarchimedean (Ind3) entry together with explicit Hodge/SHE square-weight transport data feeds such a comparison, while several simplified collapse mechanisms are rejected by Lean.

## 1 Scope

The review used the local Markdown conversions of IUT I–IV, Mochizuki’s 2026 formalization note, and the Scholze–Stix note. The relevant source passages are:

- IUT I: initial  $\Theta$ -data, prime strips, Hodge theaters, and non-scheme-theoretic  $\Theta$ -links.
- IUT II: Hodge–Arakelov evaluation, theta values of shape  $q^{j^2}$ , multiradiality, and Gaussian monoids.
- IUT III: Theorem 3.11; Corollary 3.12, especially Steps (x) and (xi); Remarks 3.12.2 and 3.12.3.
- IUT IV: later explicit log-volume estimates and Diophantine consequences.
- Scholze–Stix: the ordered real-line identification problem in their Section 2.2.
- Mochizuki 2026: the “3.11.5  $\Rightarrow$  3.12” fourth-triangle decomposition.

## 2 Mathematical Target

IUT III Step (x) treats procession-normalized log-volumes as invariants for (Ind1) and (Ind2), and turns (Ind3) into an upper inequality. Step (xi) then uses the multiradial construction, IPL, SHE, holomorphic hulls, determinants, and log-volume to place the  $q$ -pilot log-volume in an upper ray controlled by the  $\Theta$ -pilot log-volume:

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|} \quad \Rightarrow \quad -|\log(q)| \leq -|\log(\Theta)|.$$

If  $C_\Theta$  is any real number satisfying

$$-|\log(\Theta)| \leq C_\Theta |\log(q)|, \quad |\log(q)| > 0,$$

then Step (xi-f) concludes  $C_\Theta \geq -1$ . This last inference is now a Lean theorem. Mochizuki’s 2026 note isolates the final comparison as:

$$\text{Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{Corollary 3.12.}$$

The present formalization targets only this corridor, not the full construction of Theorem 3.11.

### 3 Dispute Interface

Scholze–Stix require explicit bookkeeping for all ordered one-dimensional real vector spaces. Their pressure point is that the  $\Theta$ -side concrete objects carry  $j^2$ -scaled degrees, while a consistent identification of the real lines appears to leave no room for inserting all these scalars.

The formalization therefore separates:

|                               |     |                                      |
|-------------------------------|-----|--------------------------------------|
| representative $j^2$ data     |     | raw real equality                    |
| balanced sign-compatible data |     | transported equality                 |
| aggregate averaged data       | and | matched transport scales             |
| final hull/log-volume data    |     | cancellable ordered-line comparison. |

This separation is the main correctness guard in the current code.

### 4 Lean Coverage

The relevant implementation is in

`Iut/Stage1/IUTStage1Source.lean` and `Iut/Stage1/IUTStage1Experiments.lean`.

The code currently covers:

- labeled real-line copies and positive-scale transports;
- a finite  $\mathbb{F}_\ell$ -label model with representative square weights  $j \mapsto j^2$ ;
- a representative pilot-degree profile  $\deg(\Theta_j) = j^2 \deg(q)$ ;
- proof that one label-independent scale cannot absorb all representative  $j^2$  scales in this model;
- proof that, for nonzero  $\deg(q)$ , one label-independent scale cannot account for all representative  $\Theta_j$ -degrees;
- proof that the raw representative  $j^2$  theta-degree rule does not descend to the sign quotient  $|F_\ell| = F_\ell / \{\pm 1\}$  when  $\deg(q) \neq 0$ ;
- an absolute-label exponent on  $|F_\ell| = \{0\} \cup F_\ell^\times / \{\pm 1\}$  which agrees with  $j^2$  for the canonical representatives  $0 \leq j \leq (\ell - 1)/2$ , and the corresponding theta-degree profile;
- the degree-level Gaussian evaluation  $q \mapsto (q^{j^2})_{j \in |F_\ell|}$ , including identity at  $j = 1$ ;

- a canonical representative square-weight profile on  $F_\ell$ : Lean proves the total  $j^2$ -mass is positive from the  $j = 1$  contribution and therefore constructs the profile without an external positivity premise; it also proves  $6 \sum_{j \in \mathbb{Z}/\ell\mathbb{Z}} j_{\text{rep}}^2 = (\ell - 1)\ell(2(\ell - 1) + 1)$ , hence the corresponding average formula after division by  $\ell$ ;
- the square-weighted constant-average check: a constant procession-normalized log-volume remains equal to the same constant after averaging with the canonical  $j^2$ -weights;
- the same constant-average check inside the  $F_\ell = \mathbb{Z}/\ell\mathbb{Z}$  cusp-label corridor: if the full-label log-volume is constant, the square-weighted average used in the  $\Theta$ -route is that constant, including for the canonical weights;
- the constant-target structured-SHE subcase: the formal route now proves equality of the square-weighted average with the  $\Theta$ -source average, not merely an upper bound;
- a log-volume shadow of the IUT III splitting-monoid action: the generator attached to a procession label contributes  $j^2 \deg(q)$ , contributes 0 at the core label, and adds its finite generator sum to the tensor-packet log-volume;
- the finite procession containers  $S_{j+1}^\pm = \{0, \dots, j\}$ , their nested inclusions, core-label stability, stage cardinality  $j + 1$ , and total procession ambiguity  $(\ell^\pm)!$  rather than  $(\ell^\pm)^{\ell^\pm}$ ;
- the map from the final procession container to  $|F_\ell|$ , sending the core label to 0 and recovering the exponent  $j^2$ ;
- the log-volume shadow of the tensor packet over  $S_{j+1}^\pm$ , with each factor a finite-fiber direct sum over places above  $v_\mathbb{Q}$ , total log-volume given by the finite sum over the container, normalized denominator  $j + 1$ , and invariance under container permutations;
- the  $F_\ell$ -label average specialization: for labels represented as  $\mathbb{Z}/\ell\mathbb{Z}$ , Lean rewrites the procession-normalized average denominator as  $\ell$ ;
- the IUT III Step (x) averaged-log-volume corridor: Lean records that (Ind1) and (Ind2) preserve the procession-normalized averaged log-volume, while (Ind3) supplies an upper bound;
- base-valuation products of tensor packets, the finite  $(F_{\text{mod}}^\times)_j$ -action shadow, and a coric transfer shadow  $F \rightarrow D \rightarrow D^\Gamma$  preserving product log-volume without identifying Hodge-theater histories;
- the Step (xi-d)–(xi-g) shadow: determinant normalization, upper-ray membership, and equality of the two  $q$ -pilot log-volume computations;
- the Corollary 3.12 finiteness endpoint: the statement-level value  $-|\log(\Theta)| \in \mathbb{R} \cup \{+\infty\}$  is represented by an explicit finite/ $+\infty$  split, and the hull/determinant upper-ray datum forces the finite real branch, explicitly excluding the  $+\infty$  branch; Lean also exposes the resulting real  $\Theta$ -log-volume and proves the  $q$ -pilot upper-ray comparison against that real value;
- the Corollary 3.12 statement boundary: the  $\Theta$ -pilot log-volume is typed as subject to (Ind1), (Ind2), (Ind3), while the  $q$ -pilot log-volume is typed as not subject to these indeterminacies;
- the extended statement endpoint: from the finite  $\Theta$ -branch,  $0 < -|\log(q)|$ , the upper-ray comparison, and the displayed hypothesis  $-|\log(\Theta)| \leq C_\Theta |\log(q)|$ , Lean proves  $C_\Theta \geq -1$ ;

- the Step (xi-f) final algebra: from  $-|\log(q)| \leq -|\log(\Theta)|$ ,  $|\log(q)| > 0$ , and  $-|\log(\Theta)| \leq C_\Theta |\log(q)|$ , Lean proves  $C_\Theta \geq -1$ ;
- the signed-payload bridge: the existing Stage 1 comparison record  $q_{\text{signed}} \leq \Theta_{\text{signed}}$ , together with  $\Theta_{\text{signed}} \leq C_\Theta(-q_{\text{signed}})$ , feeds this Step (xi-f) algebra directly;
- the Step (xi-g) endpoint bridge: the two-computation q-pilot record produces the signed comparison payload with  $q_{\text{signed}}$  read from the input prime-strip log-volume and  $\Theta_{\text{signed}}$  read from the hull/determinant upper-ray boundary;
- the Step (xi-e)–(xi-f) fixed-value condition: in the signed two-computation endpoint, Lean proves that the input prime-strip reading and the hull-output reading are both the same real value  $-|\log(q)|$ , and that this value lies in the upper ray  $\mathbb{R}_{\leq -|\log(\Theta)|}$ ;
- the fixed-value  $C_\Theta$  endpoint: combining the preceding fixed upper-ray membership with  $-|\log(\Theta)| \leq C_\Theta |\log(q)|$ , Lean derives  $-|\log(q)| \leq C_\Theta |\log(q)|$  and hence  $C_\Theta \geq -1$ ;
- the statement-endpoint composition: the two-computation  $C_\Theta$  endpoint canonically supplies the finite  $\Theta$ -branch Corollary 3.12 statement endpoint once the  $\Theta$ -subject/ $q$ -not-subject indeterminacy boundary is provided;
- the opening sign reduction in the proof of Corollary 3.12: from  $0 < -q_{\text{signed}}$ , Lean proves that  $0 \leq \Theta_{\text{signed}}$  immediately implies  $q_{\text{signed}} \leq \Theta_{\text{signed}}$ , matching the printed reduction to the case  $-|\log(\Theta)| < 0$ ; the same strict-negativity argument is now available at the statement endpoint for the real  $\Theta$ -log-volume;
- the Step (xi-h) warning: for  $N \geq 2$  and  $\Theta < 0$ , the tensor-power boundary satisfies  $N\Theta < \Theta$ ;
- the Step (xii) shadow: local Frobenioid  $p_v^N$ -ambiguity changes log-volume by an integer shift, while global calibration fixes exponent 0;
- the Remark 3.12.1(iii) arithmetic fact  $\mathbb{Q}_{>0} \cap \mathbb{Z}^\times = \{1\}$ ;
- the Remark 3.12.1(ii) generalized-link numerical bound  $C_\Theta \geq -\lambda$  for  $\lambda \in \mathbb{Q}_{>0}$ , with strict strengthening over the  $\lambda = 1$  bound when  $\lambda < 1$ ;
- the  $q^\lambda$ -pilot endpoint algebra: from a signed comparison for  $-\lambda |\log(q)|$  and the same  $C_\Theta |\log(q)|$  upper bound, Lean derives  $C_\Theta \geq -\lambda$ , and hence  $C_\Theta \geq -1$  when  $\lambda \leq 1$ ;
- the Remark 3.12.2 distinct-label implication  $q\text{-itw} \Rightarrow q\text{-itw} \wedge \Theta\text{-itw}/\text{indets}$ , with  $q$ -native and  $\Theta$ -native labels proved distinct;
- the Remark 3.12.2 toy calculation: forgotten concrete assignments  $-h$  and  $-2h$  are incompatible for  $h > 0$ , while  $-h \in \mathbb{R}_{\leq -2h+\epsilon}$  implies  $h \leq \epsilon$ ;
- the Fig. 3.9 column table: both  $\bullet_0$  and  $\circ$  roles are required in both columns, while  $q$ -pilot multiradiality is absent;
- the Remark 3.12.2(v) absorption distinction: zero-column unit indeterminacy is absorbed by a hull upper ray, while one-column logarithmic conjugacy is absorbed as log-volume equality;
- the Remark 3.12.3(i) metric analogy:  $\text{vol}(S) > 0$  and  $\chi_S = -\text{vol}(S)$  imply  $\chi_S < 0$ ;

- the Remark 3.12.4(iii) factor- $p$  analogy: theta-label normalization divides a gap log-volume by  $\ell$ , and multiplication by  $\ell$  recovers it;
- the Remark 3.12.4(v) p-adic analogue: an inclusion, not isomorphism, gives  $(1-p)(2g-2) \leq 0$ ;
- the IUT IV, Theorem 1.10 algebraic handoff: an explicit expression for  $C_\Theta$ , together with  $C_\Theta \geq -1$  and the remaining elementary estimates, implies the displayed  $\frac{1}{6} \log(q)$  bound;
- the final elementary coefficient estimates in IUT IV, Theorem 1.10:  $(1 - 12/\ell^2)^{-1} \leq 2$  and  $(1 - 12/\ell^2)^{-1} \frac{1}{4}(1 + 36d_{\text{mod}}/\ell) \leq 1 + 80d_{\text{mod}}/\ell$ ;
- the IUT IV, Theorem 1.10 base-change monotonicity:  $\log(d_{F_{\text{tpd}}}) + \log(f_{F_{\text{tpd}}}) \leq \log(d_F) + \log(f_F)$  propagates through the final right-hand side;
- the IUT IV, Theorem 1.10 Step (i) identities:  $n^{-1} \sum_{m=1}^n m = (n+1)/2$  and  $n^{-1} \sum_{m=1}^n m^2 = (2n+1)(n+1)/6$ ;
- the IUT IV, Theorem 1.10 Step (ii) small-prime error:  $\log(2^{11}3^35^2) \leq 21$ , represented by the corresponding linear logarithmic bound;
- the IUT IV, Theorem 1.10 Step (i)  $\text{GL}_2$  arithmetic constants for  $p = 2, 3, 5$ , including the product with the quadratic factor from (E5);
- the IUT IV, Theorem 1.10 Step (iii) algebraic estimate combining the Step (ii)  $K$ -bound and the displayed  $\log(s_Q)$ -bound into  $(1 + 4/\ell) \log(d_K, f_K) + (16/\ell) \log(s_Q) \leq (1 + 36d_{\text{mod}}/\ell)S + 2\log(\ell) + 70$ ;
- the IUT IV, Theorem 1.10 Step (viii) Proposition 1.6 case split: the two cases  $e_{\text{mod}}^* \ell \geq \eta_{\text{prm}}$  and  $e_{\text{mod}}^* \ell < \eta_{\text{prm}}$  imply the uniform  $4/3(e_{\text{mod}}^* \ell + \eta_{\text{prm}})$  bound;
- the IUT IV, Theorem 1.10 Step (viii) final absorption of  $2\log(\ell) + 74$  and the prime-product term into  $10(e_{\text{mod}}^* \ell + \eta_{\text{prm}})$ ;
- the IUT IV, Theorem 1.10 final display: the Corollary 3.12  $C_\Theta$  handoff, followed by the tripodal-to- $F$  monotonicity, gives the final  $F$ -level  $\frac{1}{6} \log(q)$  inequality;
- the IUT IV, Theorem A bounded-discrepancy conclusion: boundedness below of  $(1 + \epsilon)(\log \text{diff}_X + \log \text{cond}_D) - \text{ht}_{\omega_X}(D)$  implies the corresponding pointwise height inequality on points of bounded degree;
- the IUT IV, Corollary 2.2(i) bounded-discrepancy chain:  $\frac{1}{6} \log(q_2) \approx \frac{1}{6} \log(q_V) \approx \frac{1}{6} \text{ht}_\infty \approx \text{ht}_{\omega_X}(D)$  composes to a direct bounded discrepancy relation;
- bounded-discrepancy transfer lemmas: pointwise upper and lower bounds move across an  $\approx$  relation with the expected additive constant shift;
- the IUT IV, Corollary 2.2(ii) condition (C1) prime-scale window:  $(\log(q_V))^{1/2} \leq \ell \leq 10\delta(\log(q_V))^{1/2} \log(2\delta \log(q_V))$ . Lean keeps the square-root and logarithmic factor explicit, proves  $\log(q_V) \geq 0$ , and derives  $1 \leq 10\delta \log(2\delta \log(q_V))$  from the nonempty window;
- the IUT IV, Corollary 2.2(ii) use of (C1) in Theorem 1.10:  $\sqrt{h} \leq \ell$  and  $80d_{\text{mod}} \leq \delta$  imply  $80d_{\text{mod}}/\ell \leq \delta h^{-1/2}$ ;

- the IUT IV, Corollary 2.2(ii) error-term use of (C1):  $d_{\text{mod}}^* \leq \delta$  and  $\ell \leq 10\delta h^{1/2} \log(2\delta h)$  imply  $20(d_{\text{mod}}^* \ell + \eta_{\text{prm}}) \leq 200\delta^2 h^{1/2} \log(2\delta h) + 20\eta_{\text{prm}}$ ;
- the IUT IV, Theorem 1.10 to Corollary 2.2(ii) first-bound composition: these two C1 replacements convert the Theorem 1.10 right-hand side into  $(1 + \delta h^{-1/2})S + 200\delta^2 h^{1/2} \log(2\delta h) + 20\eta_{\text{prm}}$ ;
- the IUT IV, Corollary 2.2(ii)  $q_2/q$  comparison: the displayed estimate  $\frac{1}{6} \log(q_2) - \frac{1}{6} \log(q) \leq \frac{1}{3} h^{1/2} \log(2\delta h)$  follows through the intermediate term  $h^{1/2} \log(\ell)$ ;
- the IUT IV, Corollary 2.2(ii) pre- $\epsilon_E$  bound for  $h = \log(q_V)$ : the Theorem 1.10 bound, the  $q_2/q$  estimate, and the bounded discrepancy  $B_K$  imply

$$\frac{1}{6}h \leq (1 + \delta h^{-1/2})S + (15\delta)^2 h^{1/2} \log(2\delta h) + \frac{1}{2}C_K,$$

with  $C_K = 40\eta_{\text{prm}} + 2B_K$ ;

- the IUT IV, Corollary 2.2(ii) condition (C2) inequality chain:  $\frac{1}{6} \log(q) \leq \frac{1}{6} \log(q_2) \leq \frac{1}{6} \log(q_V) \leq (1 + \epsilon_E)(\log \text{diff}_X + \log \text{cond}_D) + C_K$ ;
- the IUT IV, Corollary 2.2(ii) definition of  $\epsilon_E$ :  $\epsilon_E = (60\delta)^2 h^{-1/2} \log(2\delta h)$ , represented with  $h^{1/2}$  and  $\log(2\delta h)$  as explicit real parameters; Lean proves positivity and  $\epsilon_E \geq 5\delta h^{-1/2}$  from  $\delta \geq 2$ ,  $h^{1/2} > 0$ , and  $\log(2\delta h) \geq 1$ ;
- the IUT IV, Corollary 2.2(ii)  $\epsilon_E$ -error conversion: substituting the definition of  $\epsilon_E$ , Lean proves  $(15\delta)^2 h^{1/2} \log(2\delta h) \leq \frac{1}{6}h \cdot \frac{2}{5}\epsilon_E$ ;
- the IUT IV, Corollary 2.2(ii) bridge to the denominator step: the pre- $\epsilon_E$   $h$ -bound, the error conversion, and  $\delta h^{-1/2} \leq \epsilon_E/5$  yield the preliminary inequality used to move the  $\frac{2}{5}\epsilon_E \frac{1}{6}h$  term to the left;
- the IUT IV, Corollary 2.2(ii) denominator step: Lean moves the term  $(2\epsilon_E/5)\frac{1}{6}h$  to the left and obtains the denominator  $(1 - 2\epsilon_E/5)^{-1}$ ;
- the IUT IV, Corollary 2.2(ii) final  $\epsilon_E$ -absorption: from  $0 < \epsilon_E \leq 1$ , Lean proves

$$\frac{1 + \epsilon_E/5}{1 - 2\epsilon_E/5} \leq 1 + \epsilon_E, \quad (1 - 2\epsilon_E/5)^{-1}C_K/2 \leq C_K,$$

and hence the displayed C2 bound from the preceding denominator expression;

- the IUT IV, Corollary 2.2(ii) final  $h$ -bound: Lean composes the pre- $\epsilon_E$  bound, error conversion, move-left step, and absorption to prove  $\frac{1}{6}h \leq (1 + \epsilon_E)S + C_K$ ;
- the IUT IV, Corollary 2.2(ii) final C2 bridge: using  $S \leq \log \text{diff}_X(x_E) + \log \text{cond}_D(x_E)$ , Lean derives the displayed C2 inequality chain;
- the IUT IV, Corollary 2.2(ii) comparison  $\log \text{diff}_X(x_E) = \log(d_{F_{\text{tpd}}})$  and  $\log(f_{F_{\text{tpd}}}) \leq \log \text{cond}_D(x_E) \leq \log(f_{F_{\text{tpd}}}) + \log(2\ell)$ ;
- the IUT IV, Corollary 2.2 to Theorem A passage: bounded-discrepancy equivalence  $\frac{1}{6} \log(q_2) \approx \text{ht}_{\omega_X(D)}$ , together with the C2 bound for  $\frac{1}{6} \log(q_2)$ , gives a lower bound for  $(1 + \epsilon_E)(\log \text{diff}_X + \log \text{cond}_D) - \text{ht}_{\omega_X(D)}$ ;

- the IUT IV, Corollary 2.2/2.3 finite-exception passage: a lower bound on the complement of a finite exceptional set and a lower bound on the exceptional set glue to a global lower bound by taking the minimum of the two constants;
- the IUT IV, Corollary 2.3 Diophantine inequality: Lean records the GenEll transfer as an explicit hypothesis and derives the global bounded-discrepancy lower bound for  $(1 + \epsilon)(\log \text{diff}_X + \log \text{cond}_D) - \text{ht}_{\omega_X}(D)$ ;
- comparison levels: pointwise representative, aggregate representative, balanced sign-compatible, and hull/log-volume;
- endpoint audits for the 3.11.5  $\Rightarrow$  3.12 comparison chart;
- nonarchimedean and archimedean (Ind3) local orientations;
- ordered real-line alignment with cancellation only after scale equality;
- a canonical unit-scale ordered alignment from a nonarchimedean (Ind3) entry;
- a bridge from factored square/full-label SHE obligations to the Hodge-descent (Ind3) route.

## 5 Current Lean Results

The current first-pass experiment report evaluates to:

|                                                          |          |
|----------------------------------------------------------|----------|
| route theorem flags                                      | = true,  |
| collapse-obstruction flags                               | = true,  |
| Gaussian/procession finite-model flags                   | = true,  |
| <b>balancedLevelRejectedAtFinalRouteTheoremAvailable</b> | = true,  |
| <b>disputeSettledByCurrentStage</b>                      | = false. |

Here the route flags are ordered real-line, nonarchimedean entry, and factored SHE availability. The obstruction flags are raw-cancellation failure, label-independent  $j^2$  failure, and sign-quotient representative descent failure. The Gaussian/procession flags include finite tensor-packet, monoid action, canonical positive square-weight profile, the constant-average check, base-valuation product, coric-transfer, determinant-normalization, upper-ray,  $q$ -pilot two-computation, the Step (x) averaged-log-volume indeterminacy corridor, tensor-power warning, and global Frobenioid-calibration shadows, plus the finiteness endpoint for  $-|\log(\Theta)| \in \mathbb{R} \cup \{+\infty\}$ , including extraction of the finite real  $\Theta$ -value, the direct Corollary 3.12 statement endpoint with  $\Theta$ -subject/ $q$ -not-subject pilot bookkeeping, the Step (xi-g)-to-(xi-f) endpoint bridge giving  $C_\Theta \geq -1$  from the two-computation  $q$ -pilot record, the Step (xi-e)-(xi-f) fixed  $-|\log(q)|$  membership check and its direct  $C_\Theta \geq -1$  consequence, the composition into the statement-level endpoint, and positive-rational unit rigidity. The generalized-link flag records the rational parameter inequality  $C_\Theta \geq -\lambda$ , now with the corresponding  $q^\lambda$ -pilot real-valued derivation. The distinct-label flag records the logical intertwining display in Remark 3.12.2 and rejects collapse of the  $q$ -native and  $\Theta$ -native labels. The toy-model flag records the resulting upper-ray arithmetic  $h \leq \epsilon$ . The column-table flag records the Fig. 3.9 theta/ $q$  similarity-difference split. The absorption flag records the zero-column upper-bound versus one-column equality distinction. The metric flag records the Gauss–Bonnet sign calculation used as analogy for the upper semi-compatibility inequality. The factor- $p$  flag records the  $F_\ell$ -label averaging denominator. The Frobenius-derivative flag records the degree inequality from inclusion. The IUT IV flag records the downstream Theorem 1.10 use of the Corollary 3.12 lower bound on  $C_\Theta$ , including the final coefficient

estimates and the tripodal-to- $F$  log-degree monotonicity. The Step (i) flag records the two elementary sum identities used in the leading-term computation. The Step (ii) small-prime flag records the additive error bound 21. The  $\text{GL}_2$  flag records the finite cardinality constants feeding that bound. The Step (iii) flag records the  $\log(s_Q)$  coefficient absorption. The Step (viii) flag records the prime-product case split used before the final upper bound for  $C_\Theta$ , plus the final coarse error absorption. The final display flag records the composed Theorem 1.10 inequality with  $F$ -level log-degree terms. The Theorem A flag records the bounded-discrepancy height/log-different/log-conductor inequality. The Corollary 2.2 flag records composition of the displayed bounded-discrepancy chain. The transfer flag records movement of pointwise bounds across bounded-discrepancy equivalence. The Corollary 2.2(C1) flag records the displayed prime-scale window and its elementary consequences. The Corollary 2.2(C2) flag records the displayed three-step logarithmic inequality chain. The C1 coefficient flag records the replacement of  $80d_{\text{mod}}/\ell$  by  $\delta h^{-1/2}$  in the Theorem 1.10-to-C2 proof. The C1 error-term flag records the corresponding replacement of  $20d_{\text{mod}}^*\ell$  by  $200\delta^2 h^{1/2} \log(2\delta h)$ . The first-bound flag records the composition of those replacements with Theorem 1.10. The  $\epsilon_E$ -definition flag records the source formula and lower estimate for  $\epsilon_E$ . The  $\epsilon_E$ -move-left flag records introduction of the denominator  $(1 - 2\epsilon_E/5)^{-1}$ . The  $\epsilon_E$ -absorption flag records the final denominator removal in C2. The  $\epsilon_E$ -error flag records the conversion of the  $(15\delta)^2$ -term to the  $\frac{2}{5}\epsilon_E \frac{1}{6}h$  term. The final  $h$ -bound flag records the full composition through absorption. The final C2 bridge flag records the last monotonic replacement of the tripodal log-degree sum by the curve different/conductor sum. The  $h$ -to-move-left flag records the bridge from the pre- $\epsilon_E$  bound to the denominator-stage preliminary inequality.  $q_2/q$  flag records the displayed comparison of  $\log(q_2)$  and  $\log(q)$ . The pre- $\epsilon_E$   $h$ -bound flag records the subsequent absorption of the bounded  $q_\vee/q_2$  discrepancy into  $C_K$ . The log-different/log-conductor flag records the bridge from tripodal field terms to curve functions. The Corollary 2.2-to-Theorem A flag records the formal bounded-discrepancy transfer from C2 to the Theorem A inequality. The finite-exception flags record the lower-bound gluing used when finite exceptional sets are enlarged in Corollary 2.2 and then passed toward Corollary 2.3. The Corollary 2.3 flag records the final GenEll-transfer interface for the Diophantine inequality. The theorem-level route is:

$$\begin{array}{ccccc}
\text{nonarch. (Ind3) entry} & \rightarrow & \text{canonical ordered } \mathbb{R}\text{-alignment} & \rightarrow & \text{source/target alignment} \\
& & \downarrow & & \downarrow \\
\text{factored SHE transport} & \rightarrow & \text{Hodge-descent audit} & \rightarrow & q_{\text{signed}} \leq \Theta_{\text{signed}}.
\end{array}$$

The route remains conditional on the displayed hypotheses.

## 6 Audit Findings

No formal statement currently proves IUT III, Corollary 3.12 from Mochizuki's published hypotheses. The strongest positive result is conditional: if the hull/log-volume upper-ray comparison supplies  $-|\log(q)| \leq -|\log(\Theta)|$ , then Lean checks the formal Step (xi-f) conclusion  $C_\Theta \geq -1$ . If the nonarchimedean (Ind3) entry, ordered real-line alignment, and Hodge/SHE square-weight transport audit are supplied, Lean checks the final signed  $q$ -to- $\Theta$  inequality feeding that algebra.

The strongest negative result is also limited: in the current  $\mathbb{F}_\ell$ -label model, Lean rejects a single label-independent scale accounting for all representative  $j^2$  factors, and it rejects balanced sign-compatible evidence as the final hull/log-volume comparison level. Lean also rejects direct descent of arbitrary raw representative  $j^2$  theta-degree data to  $|F_\ell|$ , while accepting the half-range exponent convention stated in IUT II. Lean also checks that an  $N$ -th tensor-power boundary is a strict strengthening for  $N \geq 2$  and negative pilot log-volume. It also checks the Step (xii) numerical



warning: nonzero local Frobenioid shifts change log-volume, whereas global calibration restores the unshifted value. The positive rational unit rigidity fact in Remark 3.12.1(iii) is formalized directly over  $\mathbb{Q}$ . For generalized links, Lean records that  $\lambda < 1$  makes  $-\lambda > -1$ . Lean also checks that the Remark 3.12.2 simultaneous-intertwining step depends on distinct labels rather than an unlabeled identification. In the accompanying toy model, Lean proves the final upper bound exactly from the upper-ray membership hypothesis. Lean also records that the q-pilot column lacks the theta-side multiradiality used in Theorem 3.11(iii)(c). For Remark 3.12.2(v), Lean separates hull absorption of unit shifts from log-volume equality after logarithmic-conjugate compatibility. For Remark 3.12.3(i), Lean proves the stated negative Euler-characteristic sign from positivity of metric volume. For Remark 3.12.4(iii), Lean checks the  $\ell$ -denominator normalization identity for the theta-label averaging analogy. For Remark 3.12.4(v), Lean proves the displayed nonpositive degree defect from  $p \geq 2$  and  $g \geq 2$ . For IUT IV, Theorem 1.10, Lean proves the algebraic implication from the computed  $C_\Theta$  expression and  $C_\Theta \geq -1$  to the final  $\frac{1}{6} \log(q)$  estimate, conditional on the separate elementary estimates. Lean now also proves the two displayed final elementary coefficient estimates from  $\ell \geq 5$ . It also proves the final monotonic replacement of the tripodal log-degree sum by the larger  $F$ -level log-degree sum, conditional on the two component monotonicity inequalities cited from Proposition 1.3(i). Lean also proves the Step (i) sum and square-sum identities over  $\mathbb{R}$ . For Step (ii), Lean proves the small-prime additive error bound from the displayed logarithmic inequalities. Lean also checks the Step (i)  $GL_2$  constants for  $p = 2, 3, 5$  and their product with the quadratic extension factor. For Step (iii), Lean proves the displayed coefficient absorption from the source hypotheses  $d_{\text{mod}} \geq 1$ ,  $\ell \geq 5$ ,  $2 \log(\ell) \leq \ell$ , and nonnegativity of the tripodal log-degree sum. For Step (viii), Lean checks that the two Proposition 1.6 cases imply the uniform prime-product bound used in the final  $C_\Theta$  estimate. Lean also checks the subsequent absorption into the coefficient 10 of  $e_{\text{mod}}^* \ell + \eta_{\text{prm}}$ . Combining the Corollary 3.12 handoff with the tripodal-to- $F$  monotonicity, Lean proves the final displayed  $F$ -level inequality of Theorem 1.10 at this real-valued shadow level. For Theorem A, Lean records the exact bounded-discrepancy conclusion and proves the equivalent pointwise height bound from the lower-bound hypothesis. For Corollary 2.2(i), Lean defines equality up to bounded discrepancy, proves reflexivity, symmetry, transitivity, nonnegative scaling, and composes the displayed chain from  $\log(q_2)$  to  $\text{ht}_{\omega_X(D)}$ . Lean also proves that upper and lower pointwise bounds transfer across such a bounded-discrepancy equivalence with the appropriate constant shift. For Corollary 2.2(ii)(C1), Lean records the displayed window for  $\ell$  and derives nonnegativity of  $\log(q_\forall)$ , positivity of  $(\log(q_\forall))^{1/2}$ , and  $1 \leq 10\delta \log(2\delta \log(q_\forall))$ . Lean also verifies the C1 coefficient replacement  $80d_{\text{mod}}/\ell \leq \delta h^{-1/2}$  used immediately after applying Theorem 1.10. Lean also verifies the C1 upper-window replacement for the Theorem 1.10 error term  $20(d_{\text{mod}}^* \ell + \eta_{\text{prm}})$ . Combining these, Lean proves the first displayed post-Theorem-1.10 bound in the proof of Corollary 2.2(ii). Lean also proves the displayed comparison  $\frac{1}{6} \log(q_2) - \frac{1}{6} \log(q) \leq \frac{1}{3} h^{1/2} \log(2\delta h)$  from the intermediate  $h^{1/2} \log(\ell)$  bounds. Lean then combines this with the bounded  $q_\forall/q_2$  discrepancy to prove the displayed pre- $\epsilon_E$  bound for  $\frac{1}{6}h$ . For Corollary 2.2(ii), Lean proves the final  $\frac{1}{6} \log(q)$  bound from the displayed  $\log(q)$ ,  $\log(q_2)$ , and  $\log(q_\forall)$  chain. Lean records the proof's definition of  $\epsilon_E$  and verifies the stated lower estimate  $\epsilon_E \geq 5\delta h^{-1/2}$ . Lean also verifies the source conversion  $(15\delta)^2 h^{1/2} \log(2\delta h) \leq \frac{1}{6}h \cdot \frac{2}{5}\epsilon_E$ . Combining this with  $\delta h^{-1/2} \leq \epsilon_E/5$ , Lean proves the preliminary inequality needed for the move-left step. Lean then verifies the algebraic move-left step that introduces the denominator  $(1 - 2\epsilon_E/5)^{-1}$ . Lean also proves the final  $\epsilon_E$ -absorption step used just before C2: the intermediate expression with denominator  $1 - 2\epsilon_E/5$  is bounded by  $(1 + \epsilon_E)$  times the log-degree sum plus  $C_K$ . Lean now composes these steps into the final displayed  $h = \log(q_\forall)$  bound. Lean also derives the displayed C2 chain from this  $h$ -bound and the tripodal-to-curve log-degree comparison. It also proves the two sum inequalities relating the tripodal different

and conductor terms to  $\log \text{diff}_X + \log \text{cond}_D$ . For the Corollary 2.2/2.3 exceptional-set passage, Lean proves that a lower bound off a finite exceptional set and a lower bound on the exceptional set imply a global lower bound, and specializes this to the Theorem A discrepancy. For Corollary 2.3, Lean proves the stated bounded-discrepancy conclusion from an explicit GenEll transfer hypothesis; GenEll itself is not formalized here. Finally, Lean combines the Corollary 2.2(i) bounded-discrepancy height comparison with the C2 bound to construct the Theorem A lower-bound shadow.

## 7 Missing Mathematics

The present code does not yet formalize:

- initial  $\Theta$ -data as in IUT I;
- actual Hodge theaters, Frobenioids, prime strips, log-shells, or  $\Theta$ -links;
- the Hodge–Arakelov theta evaluation and Gaussian monoids of IUT II;
- the construction of the actual theta values, Gaussian/splitting monoids,  $F_{\text{mod}}^\times$ -actions, Kummer isomorphisms, mono-analyticization, and Frobenioids behind the present shadows;
- the actual construction of log-shells and tensor packets attached to mono-analytic prime-strips in IUT III, Theorem 3.11;
- the genuine holomorphic hull and determinant operations of Step (xi), beyond the present finite determinant/upper-ray/two-computation/tensor-power shadows;
- the global realified Frobenioids of Step (xii), beyond the present integer log-volume shift model;
- the arithmetic-divisor and local log-volume construction behind the IUT IV Theorem 1.10 numerical estimates, and the Diophantine applications;
- the construction of the normalized height, log-different, and log-conductor functions used in IUT IV, Theorem A;
- the construction of the bounded-discrepancy equivalences asserted in IUT IV, Corollary 2.2(i);
- a proof that the current Hodge/SHE transport obligations follow from Mochizuki’s full hypotheses.

## 8 Next Work

The next significant formal step is not another wrapper. It is to replace current explicit Hodge/SHE transport assumptions by constructions closer to the papers:

$$\begin{aligned}
 & \text{theta values } q^{j^2} \\
 \rightsquigarrow & \text{ Gaussian and splitting monoids} \\
 \rightsquigarrow & \text{ square/full-label preservation} \\
 \rightsquigarrow & \text{ hull/log-volume route.}
 \end{aligned}$$

Only after this bridge is formalized can the first-pass experiment begin to test the actual mathematical load-bearing part of Mochizuki’s response to Scholze–Stix.

## References

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- [5] S. Mochizuki, *On the Formalization of IUT: Preliminary Progress Report*, 2026.
- [6] P. Scholze and J. Stix, *Why abc is still a conjecture*.